

1 Fubini's Theorem

Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be two finite measure spaces. Let $(X \times Y, \mathcal{F} \otimes \mathcal{G}, \mu \otimes \nu)$ be the product space. Write $\pi = \mu \otimes \nu$. Where $\mathcal{F} \otimes \mathcal{G}$ is the smallest σ -algebra generated by measurable rectangles, i.e. the Cartesian product of measurable sets of X, Y . In short,

$$\mathcal{F} \otimes \mathcal{G} = \sigma\{A \times B \mid A \in \mathcal{F}, B \in \mathcal{G}\},$$

and

$$\pi(E) = \int_X \nu(E^x) \mu(dx) = \int_Y \mu(E_y) \nu(dy).$$

Theorem 1.1 (Fubini). Let $g : X \times Y \rightarrow \overline{\mathbb{R}}$ be *product measurable*, and define

$$\begin{aligned} \alpha : X &\rightarrow \overline{\mathbb{R}}, & \alpha(x) &:= \int_Y g(x, y) \nu(dy), \\ \beta : Y &\rightarrow \overline{\mathbb{R}}, & \beta(y) &:= \int_X g(x, y) \mu(dx). \end{aligned}$$

1. If $g \geq 0$ then α and β are respectively measurable, i.e. α is $\mathcal{F}/\mathcal{B}_{\overline{\mathbb{R}}}$ -measurable and β is $\mathcal{G}/\mathcal{B}_{\overline{\mathbb{R}}}$ -measurable. Further,

$$\int_X \alpha(x) \mu(dx) = \int_Y \beta(y) \nu(dy) = \int_{X \times Y} g \, d\pi.$$

Equivalently,

$$\begin{aligned} \int_{X \times Y} g(x, y) \pi(d(x, y)) &= \int_X \int_Y g(x, y) \nu(dy) \mu(dx) \\ &= \int_Y \int_X g(x, y) \mu(dx) \nu(dy). \end{aligned} \tag{*}$$

2. For a general function $g : X \times Y \rightarrow \overline{\mathbb{R}}$, assume

$$\int_X \int_Y |g(x, y)| \nu(dy) \mu(dx) < \infty.$$

Then g is integrable. Moreover, for $x \in X$, w.r.t. μ a.s., α is well-defined and measurable, and $(*)$ holds; similarly, β is well-defined and measurable.

Proof. We proceed by measure-theoretic induction.

Step 1: Let $g = \mathbf{1}_E$, where $E \in \mathcal{F} \otimes \mathcal{G}$. Then

$$\alpha(x) = \int_Y \mathbf{1}_E(x, y) \nu(dy), \quad x \in X.$$

Fix $x \in X$. Since $\mathbf{1}_E(x, y) = \mathbf{1}_{E^x}(y)$, we have

$$\alpha(x) = \int_Y \mathbf{1}_{E^x}(y) \nu(dy) = \nu(E^x), \quad \forall x \in X.$$

Hence, by the measurability result for sections, $\alpha : X \rightarrow \overline{\mathbb{R}}$ is $\mathcal{F}/\mathcal{B}_{\overline{\mathbb{R}}}$ -measurable.

Similarly,

$$\beta(y) = \int_X \mathbf{1}_E(x, y) \mu(dx) = \int_X \mathbf{1}_{E_y}(x) \mu(dx) = \mu(E_y), \quad \forall y \in Y,$$

and therefore $\beta : Y \rightarrow \overline{\mathbb{R}}$ is $\mathcal{G}/\mathcal{B}_{\overline{\mathbb{R}}}$ -measurable.

Further, since $\pi = \mu \otimes \nu$ by the definition of the product measure,

$$\pi(E) = \int_X \alpha \, d\mu = \int_Y \beta \, d\nu,$$

which proves (*) and (**) for indicator functions.

Step 2: Suppose (*) and (**) hold for $g_1, g_2, \dots, g_n \geq 0$ which are product measurable, and let $a_1, a_2, \dots, a_n \geq 0$. Set $g = \sum_{i=1}^n a_i g_i$. Then, by Step 1 and linearity of the integral,

$$\begin{aligned} \int_X \int_Y g(x, y) \, \nu(dy) \, \mu(dx) &= \int_X \int_Y \sum_{i=1}^n a_i g_i(x, y) \, \nu(dy) \, \mu(dx) \\ &= \sum_{i=1}^n a_i \int_X \int_Y g_i(x, y) \, \nu(dy) \, \mu(dx), \end{aligned}$$

and similarly for the other order of integration. Hence (*) and (**) hold for every nonnegative measurable simple function.

Step 3: Suppose (*) and (**) hold for $\{g_n\}_{n \geq 1}$ here each is nonnegative and measurable, and $g_n \uparrow g$. Then, for every fixed ,

$$g_n(x, y) \uparrow g(x, y).$$

Hence, by the Monotone Convergence Theorem, for every $x \in X$,

$$\int_Y g_n(x, y) \, \nu(dy) \uparrow \int_Y g(x, y) \, \nu(dy),$$

and, for every $y \in Y$,

$$\int_X g_n(x, y) \, \mu(dx) \uparrow \int_X g(x, y) \, \mu(dx).$$

Applying MCT once more in the outer integral gives

$$\int_X \int_Y g_n(x, y) \, \nu(dy) \, \mu(dx) \uparrow \int_X \int_Y g(x, y) \, \nu(dy) \, \mu(dx),$$

and similarly,

$$\int_Y \int_X g_n(x, y) \, \mu(dx) \, \nu(dy) \uparrow \int_Y \int_X g(x, y) \, \mu(dx) \, \nu(dy).$$

Since (*) and (**) hold for every g_n , they also hold for g .

Thus (*) and (**) hold for every nonnegative measurable function by the monotone approximation theorem. $\square$

Part 2. By Part 1 and the assumption, we get

$$\int_{X \times Y} |g| \, d\pi < \infty \implies g \text{ is integrable.}$$

Further, by Part 1,

$$\int_X \left(\int_Y |g(x, y)| \, \nu(dy) \right) \mu(dx) < \infty \implies \int_Y |g(x, y)| \, \nu(dy) < \infty \quad \mu\text{-a.e. } x \in X$$

We write

$$g = g^+ - g^-, \quad |g| = g^+ + g^-,$$

where

$$g^+ := \max\{g, 0\}, \quad g^- := \max\{-g, 0\}.$$

Thus, for μ -a.e. x ,

$$\int_Y g^+(x, y) \nu(dy) < \infty, \quad \int_Y g^-(x, y) \nu(dy) < \infty.$$

Therefore

$$\alpha(x) := \int_Y g(x, y) \nu(dy)$$

is well-defined and finite for μ -a.e. x .

Moreover,

$$\int_X \int_Y g^+(x, y) \nu(dy) \mu(dx) = \int_{X \times Y} g^+ d\pi < \infty,$$

and similarly,

$$\int_X \int_Y g^-(x, y) \nu(dy) \mu(dx) = \int_{X \times Y} g^- d\pi < \infty.$$

Hence, by subtracting,

$$\begin{aligned} \int_X \alpha d\mu &= \int_X \int_Y g(x, y) \nu(dy) \mu(dx) \\ &= \int_{X \times Y} g d\pi. \end{aligned}$$

Similarly, working with

$$\beta(y) := \int_X g(x, y) \mu(dx),$$

we obtain

$$\int_X g^+(x, y) \mu(dx) < \infty, \quad \int_X g^-(x, y) \mu(dx) < \infty \quad \nu\text{-a.e. } y.$$

Thus β is well-defined and finite for ν -a.e. y , and

$$\begin{aligned} \int_Y \beta d\nu &= \int_Y \int_X g(x, y) \mu(dx) \nu(dy) \\ &= \int_{X \times Y} g d\pi. \end{aligned}$$

Consequently,

$$\boxed{\int_X \int_Y g(x, y) \nu(dy) \mu(dx) = \int_{X \times Y} g d\pi = \int_Y \int_X g(x, y) \mu(dx) \nu(dy)}.$$

Ba dum tsssss.